

## Bachelor in Computer Applications

Course Title: **Database Management System**

Code No:    Semester: **3**

Full Marks: **60**    Pass Marks: **24**    Time: **3 hours**

*Candidates are required to answer the questions in their own words as far as practicable.*

**Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

### Group B

1. What is an E-R Diagram? Explain various symbols used in the E-R Diagram.
2. What is Transaction? State and explain the properties of Transaction.
3. From the relation given below, answer the following questions.  
EMPLOYEE (Ssn, Fname, Minit, Name, Bdate, Address, Sex, Salary, Super\_ssn, Dno)  
DEPARTMENT (Dname, Dnumber, Mgr\_ssn, Mgr\_start\_date)  
PROJECT (Pname, Pnumber, Plocation, Dnum)  
DEPENDENT (Essn, Dependent\_name, Sex, Bdate, Relationship)  
DEPT\_LOCATIONS (Dnumber, Dlocation)  
a) Retrieve all the attributes of employees who work for the 'Computer' department department no 5 using SQL  
b) For each Department, retrieve the department number, the number of employees in department, and their average salary using SQL  
c) For each project in which more than two employees work, retrieve the project number, name, and the number of employees who work on that project using SQL  
d) Select the EMPLOYEE tuples whose department is 10, or those whose salary is greater than Rs 50,000 using Relational Algebra.  
e) Retrieve the names of employees who have no dependents using Relational Algebra
4. What are different types of Database Users? Explain in Brief.
5. What is Primary key and Foreign Key? Why is this required in the DBMS?
6. What is Integrity constraint? What are the three main categories of Integrity constraints?
7. What is Functional Dependency? Explain 3NF with Examples.
8. What is Schedule? Explain Serializability and Conflict Schedule.
9. What is starving in DBMS? Explain with example.
10. During ER-to-Relational Mapping, explain with example to map 1: N and N:M relationship into a relation?
11. What is NoSQL? What are the types of NoSQL? Explain.
12. Explain Data Models, Schemas, and Instances.